Chapter 1

Soil: The Ground Beneath Our Feet

Soil is the top layer of the Earth in which plants grow. We get our food, cotton for our clothes and wood for our houses from plants that grow in soil.

How soil is made

Soil is made from rocks. Over a very, very long time, the heat of the Sun, the wind and the rain break rocks into smaller and smaller pieces. These tiny bits of rock become part of the soil. It can take hundreds of years to make just a few centimetres of soil.

When leaves, plants and dead animals rot, they mix with the soil. This rotting matter is called humus. Humus is dark in colour and makes the soil rich, so that plants grow well in it.

Life in the soil

Soil is full of living things. Earthworms, ants and many tiny creatures live in it. Earthworms make tunnels that let air and water go deep into the soil. They also eat dead leaves and make the soil rich. That is why earthworms are called the friends of farmers.

Not all soil is the same

If you collect soil from different places, you will find that it has different colours. Some soil is dark, some is red and some is light brown. Some soil feels soft and smooth, and some feels rough and sandy.

Types of soil in India

India has many types of soil. Four of them are:

- Alluvial soil is brought by rivers and spread on the land around them. It is found in the large plains of the north where the Ganga flows. It is very fertile, and wheat and rice grow well in it.
- Black soil is dark in colour and holds water for a long time. Cotton grows very well in it. It is found in Maharashtra and Gujarat.
- Red soil looks red because it has iron in it. It is found in Tamil Nadu, Karnataka and Odisha.
- Desert soil is sandy and cannot hold water. It is found in Rajasthan.

Protecting our soil

Strong winds and heavy rain can carry away the top layer of soil. This is called soil erosion. When soil is washed away, plants cannot grow well.

The roots of trees and grass hold the soil together. So, planting trees is the best way to protect our soil.

Let's think

1. How is soil made?
2. Why are earthworms called the friends of farmers?
3. Name the four types of soil in India given in this chapter.

New words

humus - the rich, dark part of soil made from rotting leaves and plants
fertile - able to grow good crops